

TØ 10 – Mikroøkonomi 2 – 2024

Problem 1

A risk neutral entrepreneur has two projects available, each requiring an investment of 10 at $t=0$. The two projects are mutually exclusive, i.e. he must choose one of them. At $t=1$ the first project generates a payment of 48 with probability a , or 8 with probability $(1-a)$. At $t=1$ the second project generates a payment of 55 with probability a , or 0 with probability $(1-a)$.

The probability a of the high payment is also the level of effort chosen by the entrepreneur. $a \in \{0, \frac{1}{3}, \frac{1}{2}\}$. The entrepreneur has a cost of effort equal to $40a^2$, and there is no discounting.

1. If the entrepreneur pays the initial investment himself, what level of effort will he choose under each project? In which project will he invest? Hint: for each project determine which effort, a , that gives the entrepreneur the highest expected payment.

Suppose that the entrepreneur finances the investment by issuing debt. A debt contract is an agreement which secures the creditor a cashflow of $\min\{D, C_i\}$ in return for his loan of 10 at $t = 0$. C_i refers to the *realized* cash flow at $t = 1$ (i.e. for project 1 either 8 or 48). D is referred to as the face value of the debt. Assume that the creditor is risk neutral and observes which project is undertaken. Assume the credit market is fully competitive, i.e. expected profits for the creditor are 0.

2. For each project write up an equation which determines D when expected revenues are equal to the initial loan for the creditor (this is the (binding) PC condition for the creditor).

1

1 symbolises the first project and 2 symbolises the second project

At $t = 0$:

1: -10

2: -10

At $t = 1$

1: 48 with probability a or 8 with probability $(1-a)$

2: 55 with probability a or 0 with probability $(1-a)$

$$a \in \left\{0, \frac{1}{3}, \frac{1}{2}\right\}$$

The cost of effort is

$$c = 40a^2$$

I find out which level of effort the entrepreneur will choose under each project by first determining with level of effort, a , that gives him the highest expected payment.

General formula is

$$E[I_j] = -10 + P_H \cdot a + P_L \cdot (1 - a) - 40a^2$$

Where P_H (P_L) is high (low) payment

With the first project

$$E[I_1] = -10 + 48 \cdot 0 + 8(1 - 0) - 40 \cdot 0^2 = -2, \quad a = 0$$

$$E[I_1]' = -10 + 48 \cdot \frac{1}{3} + 8 \left(1 - \frac{1}{3}\right) - 40 \cdot \left(\frac{1}{3}\right)^2 = 6.88, \quad a = \frac{1}{3}$$

$$E[I_1]'' = -10 + 48 \cdot \frac{1}{2} + 8 \left(1 - \frac{1}{2}\right) - 40 \cdot \left(\frac{1}{2}\right)^2 = 8, \quad a = \frac{1}{2}$$

Implying that in the first project, the entrepreneur will choose effort level $a = \frac{1}{2}$

With the second project

$$E[I_2] = -10 + 55 \cdot 0 + 0(1 - 0) - 40 \cdot 0^2 = -10, \quad a = 0$$

$$E[I_2]' = -10 + 55 \cdot \frac{1}{3} + 0 \cdot \left(1 - \frac{1}{3}\right) - 40 \cdot \left(\frac{1}{3}\right)^2 = 3.89, \quad a = \frac{1}{3}$$

$$E[I_2]'' = -10 + 55 \cdot \frac{1}{2} + 0 \cdot \left(1 - \frac{1}{2}\right) - 40 \cdot \left(\frac{1}{2}\right)^2 = 7.5, \quad a = \frac{1}{2}$$

Implying that in the second project, the entrepreneur will choose effort level $a = \frac{1}{2}$

Because $E[I_1]'' > E[I_2]''$, the entrepreneur will choose the first project

2

At $t = 0$:

Cashflow of $\min\{D, C_i\}$ where C_i is the realised cash flow at $t = 1$ and D is the face value of the debt. When the high (low) profit is reached, it is greater (lower) than the face value of debt, i.e., $8 < D < 48$

For the first project

$$a \cdot \min\{D, 48\} + (1 - a) \min\{D, 8\} - 10 = a \cdot D + (1 - a) \cdot 8 = \bar{u} = 0$$

For the second project

$$a \cdot \min\{D, 55\} + (1 - a) \min\{D, 0\} - 10 = a \cdot D + (1 - a) \cdot 0 = aD = \bar{u} = 0$$

3. What level of effort will the entrepreneur choose under each project? Solve for D under each project and determine which project the entrepreneur will invest in.

Hint: For each project remember to include the equation from part 2 as a constraint.

4. Now assume that the issuer of debt does not observe which project the entrepreneur invests in. Therefore a debt contract is unconditional on which project is chosen. Which project will the entrepreneur choose? What will the bank do in response? Calculate the value of D in the market.

3

In the first project, the entrepreneur solves

$$\begin{aligned} a \cdot (48 - D) + (1 - a)(8 - 8) - 40a^2 \\ \text{s.t. } a \cdot D + (1 - a) \cdot 8 = \bar{u} \end{aligned}$$

I rewrite the PC as

$$a \cdot D + (1 - a) \cdot 8 - 10 = 0 \Leftrightarrow D = \frac{8a + 2}{a}$$

And now I rewrite the maximisation problem

$$\begin{aligned} a \cdot (48 - D) + (1 - a)(8 - 8) - 40a^2 &= a \cdot \left(48 - \frac{8a + 2}{a}\right) - 40a^2 \\ &= 48a - 8a - 2 - 40a^2 \end{aligned}$$

FOC

$$48 - 8 - 80a = 0 \Leftrightarrow a^* = \frac{1}{2}$$

Face value of debt

$$D = \frac{8a + 2}{a} = 12$$

Entrepreneur's profit is

$$a \cdot (48 - D) + (1 - a)(8 - 8) - 40a^2 = a(48 - D) - 40a^2 = \frac{1}{2}(48 - 12) - 10 = 8$$

Second project

$$a \cdot (55 - D) - 40a^2$$
$$s.t. a \cdot D = 10$$

I rewrite the PC

$$D = \frac{10}{a}$$

And now I rewrite the maximisation problem

$$a \cdot (55 - D) - 40a^2 = a \left(55 - \frac{10}{a} \right) - 40a^2 = 55a - 10 - 40a^2$$

FOC

$$55 - 80a = 0 \Leftrightarrow a = \frac{11}{16}$$

Which is not a solution, so the one closest to $a = \frac{11}{16}$ is chosen, i.e., $a = \frac{1}{2}$

Face value

$$D = \frac{10}{a} = 20$$

Profit

$$a \cdot (55 - D) - 40a^2 = \frac{1}{2} \cdot (55 - 20) - 10 = 7.5$$

Therefore, project 1 is chosen

4

Entrepreneur will always choose 2 since

$$a \cdot (48 - \min\{D, C_i\}) + (1 - a) \cdot (8 - \min\{D, C_i\})$$
$$a(48 - D) + (1 - a)(8 - 8) < a(55 - D) \forall a > 0$$

The bank will then anticipate project 2 is and hence require $D = 20$ since the PC is (with $a^* = \frac{1}{2}$):

$$\frac{1}{2}D = 10 \Leftrightarrow D = 20$$

Problem 2

An employer (P - principal) is looking for a new employee (A - agent) to work for him in his firm. Due to a Corona lockdown the employee will be working from home and the employer will therefore not directly observe the employee's activity a . Instead P will observe the total sales of his firm determined as $q = a + \epsilon$ where ϵ is a normally distributed random variable with mean 0 and variance σ^2 . Due to unions and boring legal stuff, the employer is further only able to offer linear contracts of the form $w = t + sq$ where t is a base amount and s governs how close the wage of the employee is linked to firm performance. The employer is risk neutral and interested in maximizing the expected profits of his firm.

The agent has CARA preferences represented by

$$u(w, a) = -e^{-\eta(w - \psi(a))}$$

where w is the wage the agent gets when working and $\eta > 0$ is the agent's coefficient of absolute risk aversion. $\psi(a)$ gives the cost of being active at level a and is given by $\psi(a) = \frac{1}{2}ca^2$ where $c > 0$. The outside option of the agent is given by

$$u(\bar{w}) = -e^{-\eta(\bar{w})}$$

1. What type of problem is this? Is there a conflict of interest and scope for postcontractual opportunism?
2. For a given (accepted) contract (t, s) formulate the maximization problem of the agent when deciding on the level of a .

Hint: The agent is maximizing expected utility. When ϵ is normally distributed with zero mean and variance σ^2 we have that for any γ the expected value is $E(e^{\gamma\epsilon}) = e^{\frac{\gamma^2\sigma^2}{2}}$.

3. Solve for the level of a chosen by the agent, i.e. derive the first order conditions and solve for an expression for a .

1

Moral Hazard type of problem: Ceteris paribus, an increase in the employee's activity, a , results in a lower utility for the agent but a higher profit for principal.

2

$$w = t + sq$$

$$\psi(a) = \frac{1}{2}ca^2$$

$$q = a + \epsilon$$

Maximisation problem is

$$\begin{aligned} \max \mathbb{E}[-e^{-\eta(w - \psi(a))}] &= \max \mathbb{E}\left[-e^{-\eta\left(t + sq - \frac{1}{2}ca^2\right)}\right] \\ &= \max \mathbb{E}\left[-e^{-\eta\left(t + s(a + \epsilon) - \frac{1}{2}ca^2\right)}\right] \end{aligned}$$

$$\begin{aligned}
&= \max -e^{-\eta\left(t+sa-\frac{1}{2}ca^2\right)} \cdot \mathbb{E}[e^{-\eta s\varepsilon}] \\
&= \max -e^{-\eta\left(t+sa-\frac{1}{2}ca^2\right)} \cdot e^{\frac{(\eta s)^2 \sigma^2}{2}} \\
&= \max -e^{-\eta\left(t+sa-\frac{1}{2}ca^2\right)-\frac{(\eta s)^2 \sigma^2}{2}} \\
&= \max -e^{-\eta\left(t+sa-\frac{1}{2}ca^2-\frac{\eta \cdot s^2 \sigma^2}{2}\right)}
\end{aligned}$$

Where I have used the hint: $\mathbb{E}[e^{\eta s \cdot \varepsilon}] = e^{\frac{(\eta s)^2 \sigma^2}{2}}$

3

The FOC from the maximisation problem in part 2 is the partial derivative of the expected utility w.r.t. a :

$$\frac{\partial}{\partial a} \left(-e^{-\eta\left(t+sa-\frac{1}{2}ca^2-\frac{1}{2}\eta s^2 \sigma^2\right)} \right) = -\eta \left(-e^{-\eta\left(t+sa-\frac{1}{2}ca^2-\frac{1}{2}\eta s^2 \sigma^2\right)} \right) (s - ca) = 0$$

Which is satisfied when $(s - ca) = 0 \Leftrightarrow a = \frac{s}{c}$ since $-\eta \left(-e^{-\eta\left(t+sa-\frac{1}{2}ca^2-\frac{1}{2}\eta s^2 \sigma^2\right)} \right) \neq 0$

Because $\eta > 0$ and $-e^{-\eta\left(t+sa-\frac{1}{2}ca^2-\frac{1}{2}\eta s^2 \sigma^2\right)} \neq 0$ by definition

4. Having solved for a for a given contract (t, s) , formulate the principal's problem under the second best.

Hint: Instead of an IC constraint use the expression for a from above.

5. Argue that the PC constraint is binding under the second best, and use this to formulate the constraint as

$$\bar{w} = t + \frac{s^2}{c} - \frac{c}{2} \frac{s^2}{c^2} - \frac{\eta}{2} s^2 \sigma^2$$

6. Derive the two first order conditions and interpret them.

4

Since P knows that A chooses $a = \frac{s}{c}$, maximisation problem is now (remember $q = a + \varepsilon$)

$$\begin{aligned}
\max \mathbb{E}[q - w] &= \mathbb{E}[a + \varepsilon - w] \\
&= \mathbb{E}[a + \varepsilon - (t + sq)]
\end{aligned}$$

$$\begin{aligned}
&= \frac{s}{c} + \mathbb{E}[\varepsilon] - \mathbb{E}[t + sq] \\
&= \frac{s}{c} - (t + \mathbb{E}[sq]) \\
&= \frac{s}{c} - (t + \mathbb{E}[(a + \varepsilon) \cdot s]) \\
&= \frac{s}{c} - (t + \mathbb{E}[a \cdot s]) \\
&= \frac{s}{c} - (t + a \cdot \mathbb{E}[s]) \\
&= \frac{s}{c} - \left(t + \frac{s}{c} \cdot \mathbb{E}[s]\right) \\
&= \frac{s}{c} - \left(t + \frac{s}{c} \cdot s\right) \\
&= \frac{s}{c} - \left(t + \frac{1}{c} s^2\right)
\end{aligned}$$

$s. t.$

$$-e^{-\eta\left(t+sa-\frac{1}{2}ca^2-\frac{1}{2}\eta s^2\sigma^2\right)} = -e^{-\eta\left(t+\frac{s^2}{c}-\frac{1}{2}cs^2-\frac{1}{2}\eta s^2\sigma^2\right)} \geq -e^{-\eta(\bar{w})}$$

The constraint is a PC which at the same times satisfies IC by setting $a = \frac{s}{c}$ as the hint suggests

5 + 6

PC is binding since otherwise, the wage could be lowered while still being greater than or equal to the reservation utility

Lagrange

$$\mathcal{L} = a - \left(t + \frac{1}{c}s^2\right) + \lambda \bar{w} = \frac{s}{c} - \left(t + \frac{1}{c}s^2\right) + \lambda \left(t + \frac{1}{c}s^2 - \frac{1}{2}c \cdot \frac{1}{c^2}s^2 - \frac{1}{2}\eta s^2\sigma^2\right)$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial t} = -1 + \lambda = 0 \Leftrightarrow \lambda = 1$$

$$\frac{\partial \mathcal{L}}{\partial s} = \frac{1}{c} - \frac{2s}{c} + \frac{2\lambda s}{c} - \frac{sc\lambda}{c^2} - \lambda\eta s\sigma^2 = 0 \Leftrightarrow s = \frac{1}{1 + c\eta\sigma^2}$$

Interpretation of the FOC: s is a measure of how close the wage of the employee is linked to firm performance, so, *ceteris paribus*, this so-called correlation between the agent's wage and the firm's performance—a performance pay—will decrease if either the agent becomes more risk-averse (η), the cost of effort increases (c), or if the variance becomes larger as a sort of signal jamming will appear.

Problem 3:

Consider the following market for used cars. There are many sellers of used cars. Each seller has exactly one used car to sell and is characterized by the quality of the used car he wishes to sell. Let $\theta \in [0, 1]$ index the quality of a used car and assume that is uniformly distributed on $[0, 1]$. If a seller sells his car of quality θ for a price p , his utility is $u_s(p, \theta)$. If he does not sell a car his utility is 0.

Buyers of used cars receive utility $\theta - p$ if they buy a car of quality θ at price p and receive utility 0 if they do not purchase a car.

There is asymmetric information regarding the quality of used cars. Sellers know the quality of the car they are selling, but buyers do not know the quality. Assume that there are not enough cars to supply all potential buyers.

- (a) Determine the equilibrium price(s) if $u_s(p, \theta) = p - \theta^3$. Describe an equilibrium in words. In particular which cars are traded in equilibrium? (write at most 3 sentences)
- (b) Determine the equilibrium price(s) if $u_s(p, \theta) = p - \sqrt{\theta}$.
- (c) Determine the equilibrium price(s) if $u_s(p, \theta) = p - \frac{\theta}{2}$.
- (d) Are the outcomes in 1, 2 and 3 Pareto efficient? Describe whether some equilibria are “closer to Pareto efficiency” than others (Pareto improvements) whenever possible.

q is the same as θ . I just use the notation from the slides instead

1

The quality of used cars is uniformly distributed with: $q \in [0, 1]$

Utility of selling a car is $u_s = u_s(p, q)$

Otherwise: 0

Utility of buying a car is $u = q - p$

Otherwise: 0

Sellers will sell at price (i.e., the expected price conditional on the PC):

$$p = \mathbb{E}[p | u_s(p, q) \geq 0]$$

2

Expected valuation of a car bought at a given price p is

$$E[q | p - q^3 \geq 0] = E\left[q \mid p^{\frac{1}{3}} \geq q\right]$$

$$q = \frac{\int_0^{p^{\frac{1}{3}}} q dq}{p^{\frac{1}{3}}} = \frac{\left[\frac{1}{2} q^2\right]_0^{p^{\frac{1}{3}}}}{p^{\frac{1}{3}}} = \frac{1}{2} p^{\frac{1}{3}}$$

Now, setting the expected utility equal to zero yields

$$E[U] = q - p = \frac{1}{2} p^{\frac{1}{3}} - p = 0 \Leftrightarrow p \in \left\{0, \frac{1}{4} \sqrt{2}\right\}$$

Which means that cars with a quality of

$$q \leq \frac{1}{4} \sqrt{2}$$

will get sold. That is because this condition must hold

$$q \leq p$$

3

Pretty much the same task as part 2, only now, the utility is $u_s(p, q) = p - q^{\frac{1}{2}}$

$$E\left[q \mid p - q^{\frac{1}{2}} \geq 0\right] = E[q | p^2 \geq q]$$

$$q = \frac{\int_0^{p^2} q dq}{p^2} = \frac{1}{2} p^2$$

$$E[U]' = q - p = \frac{1}{2} p^2 - p$$

Now, setting the expected utility equal to zero yields

$$E[U]' = \frac{1}{2} p^2 - p = 0 \Leftrightarrow p \in \{0, 2\}$$

2 is not an equilibrium since no buyer will buy at price $p = 2$

4

Now, the utility is $u_s(p, q) = p - \frac{1}{2} q$

$$E \left[q \left| p - \frac{1}{2}q \geq 0 \right. \right] = E[q | 2p \geq q]$$

$$q = \frac{\int_0^{2p} q dq}{2p} = p$$

I calculate the expected utility

$$E[U]'' = q - p = p - p = 0$$

Which means that every car will get sold and every $p \in \left[0, \frac{1}{2}\right]$ is an equilibrium since the inequality $p - \frac{1}{2}q \geq 0$ must hold where $q \in [0,1]$.

The equilibrium implies that the higher the price, the more cars are being sold

5

2

Buyers' valuation exceeds the sellers and therefore, every car is sold

3

Opposite scenario here: the sellers' valuation exceeds the buyers', and no cars are hence sold.

4

When $p = \frac{1}{2}$, price is maxed such that every car is sold, and it is thus efficient if and only if

$p = \frac{1}{2}$ since Pareto efficiency requires that every car is sold